Task 1. Install Docker and verify installation

For Windows, install Docker Desktop.

- Download: Docker Desktop

After installation, open PowerShell and execute:

- docker --version

Check whether Docker is running:

- docker info

Task 2. Download Nginx image

Pull the official Nginx image:

- docker pull nginx

Display downloaded images:

- docker images

Task 3. Run an Nginx container

Execute:

- docker run -d -p 8080:80 --name mynginx nginx

Check the running container:

- docker ps

Task 4. Create and attach a Docker volume

Create a volume:

- docker volume create zomato_data

Check the volume:

- docker volume ls

Run Apache with the volume:

- docker run -d -p 8081:80 --name zomato-apache -v zomato_data:/usr/local/apache2/htdocs httpd

Verify:

- docker ps

Test Apache:

- Open Apache Server

To verify persistent storage, create a file inside the container:

- docker exec zomato-apache sh -c "echo 'Welcome to Zomato' > /usr/local/apache2/htdocs/index.html"

Open the browser again:

- http://localhost:8081

You should see:

- Welcome to Zomato

Even if you remove the container, the volume remains:

- docker rm -f zomato-apache
- docker volume ls

Task 5. Three benefits of Docker over VMs

1. Faster deployment: Zomato can start new containers quickly during high food-order traffic.
2. Resource efficiency: Instagram can run multiple containers without installing a separate operating system for each application.
3. Portability: Flipkart can deploy the same Docker image on local servers, AWS, or other cloud platforms.